

Blood Cell Image Generation: A Comparative Study of Deep Generative Models

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Abstract. This work presents a comparative analysis of four generative models—DCGAN, VAE, DDPM, and PixelCNN—for blood cell image synthesis using the BloodMNIST dataset. Using hyperparameter optimization and evaluation with different random seeds, we assessed model performance through Fréchet Inception Distance (FID). Our results indicate that DDPM achieves the highest quality synthetic images with an average FID score of 38.60, significantly outperforming VAE (148.08), DCGAN (129.46), and PixelCNN (234.24). We discuss the trade-offs between model complexity, training stability, and computational requirements, providing insights into suitable generative approaches for medical image synthesis.

Keywords: Generative models · DCGAN · VAE · DDPM · Medical imaging · Blood cell images

1 Introduction

Generative models have advanced significantly, enabling synthesis of realistic data across various domains. In medical imaging, these models offer solutions to challenges like data scarcity and class imbalance by producing synthetic yet realistic images [9]. The BloodMNIST dataset [8] provides a valuable resource for hematological analysis, containing labeled images of eight different blood cell types.

This study compares four distinct generative approaches—DCGAN [5], VAE [2], DDPM [3], and PixelCNN [4]—for blood cell image synthesis. Each represents a different paradigm: adversarial training (DCGAN), variational inference (VAE), diffusion processes (DDPM), and autoregressive modeling (PixelCNN). By systematically evaluating these models, we aim to identify the most effective techniques for medical image generation.

2 Background

DCGAN: Deep Convolutional Generative Adversarial Networks extend the GAN framework [1] with convolutional architectures. GANs operate through a two-player game where the generator creates samples to fool the discriminator, while the discriminator distinguishes real from generated samples [5].

VAE: Variational Autoencoders [2] combine variational inference with neural networks to learn latent representations. VAEs impose a prior distribution on the latent space and optimize an evidence lower bound that balances reconstruction quality against KL divergence.

DDPM: Denoising Diffusion Probabilistic Models [3] gradually convert noise into data through iterative denoising. During training, the model learns to predict noise added at each step, enabling generation by progressively denoising random noise.

PixelCNN: PixelCNN [4] generates images pixel by pixel sequentially. Each pixel’s distribution is conditioned on previously generated pixels, using masked convolutions to maintain autoregressive generation.

3 Methodology

3.1 Dataset and Model Implementations

The BloodMNIST dataset [8] consists of 17,092 RGB images (28×28 pixels) of eight blood cell types. We combined all splits to maximize training data and normalized images appropriately for each model.

For each model, we implemented architectures adapted for BloodMNIST:

DCGAN: Generator with transposed convolutions; discriminator with convolutional layers, leaky ReLU and batch normalization.

VAE: Encoder mapping to latent distributions; decoder reconstructing from latent samples; loss balancing reconstruction and KL divergence.

DDPM: U-Net architecture with time embeddings, linear noise scheduling over 1,000 timesteps.

PixelCNN: Masked convolutions with gated activation units and residual connections.

3.2 Hyperparameter Optimization and Evaluation

We performed grid search over key hyperparameters:

DCGAN: Latent dimension [100], feature maps [64, 128], learning rates [0.0002, 0.0001], beta1 [0.5, 0.0]

VAE: Hidden dimensions [[512, 256], [1024, 512, 256], [2048, 1024, 512]], latent dimension [32, 128], [0.1, 0.5, 1.0]

DDPM: Model channels [64, 128], time embedding [256], learning rates [0.0001, 0.0002, 0.0005]

PixelCNN: Hidden dimensions [64, 96], residual blocks [3, 5], learning rates [0.001, 0.0005], weight decay [1e-5]

All models were trained using Adam optimizer with early stopping. We evaluated using Fréchet Inception Distance (FID) between 10,000 real and 10,000 generated images, with five runs per model using different random seeds.

4 Results

4.1 Best Hyperparameter Configurations

Table 1. Best hyperparameter configurations for each model.

Parameter	DCGAN	VAE	DDPM	PixelCNN
Latent dimension	100	128	—	—
Hidden dimensions	—	[512, 256]	—	64
Generator/Model channels	128	—	128	—
Discriminator channels	128	—	—	—
Time embedding	—	—	256	—
Residual blocks	—	—	—	3
Learning rate	0.0001	0.0003	0.0002	0.001
Beta1	0.5	—	—	—
Beta (KL weight)	—	0.1	—	—
Weight decay	—	—	—	1e-5
Best FID Score	129.46	148.08	38.60	152.10

4.2 Quantitative Results

Table 2 presents average FID scores across five runs. DDPM significantly outperformed other models, achieving an FID score nearly four times better than VAE, three times better than DCGAN, and six times better than PixelCNN. It’s worth noting the high standard deviation in DCGAN results, indicating significant variability in performance across different random seeds.

Table 2. Average FID scores across five runs with different random seeds.

Model	Average FID Score	Standard Deviation
DCGAN	246.38	± 103.14
VAE	148.08	± 2.30
DDPM	38.60	± 1.43
PixelCNN	234.24	± 16.39

4.3 Qualitative Model Analysis

DCGAN Despite achieving satisfactory performance in its best run (FID 129.46), DCGAN exhibited severe inconsistency across different random seeds. The high standard deviation in FID scores (± 103.14) and frequent mode collapse resulting in entirely pink images highlights a significant limitation of adversarial approaches. This training instability makes DCGANs unreliable for medical applications where consistency is critical.

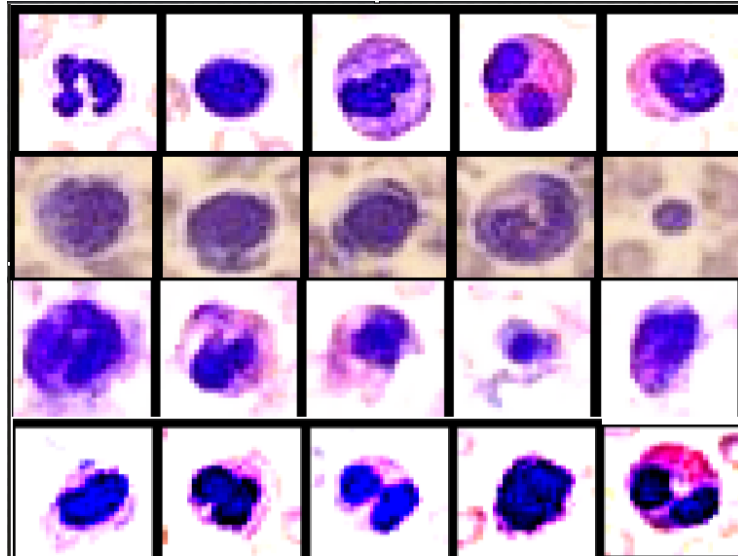


Fig. 1. Comparison of real BloodMNIST images (top row) with images generated by DCGAN (second row), VAE (third row), and DDPM (bottom row).

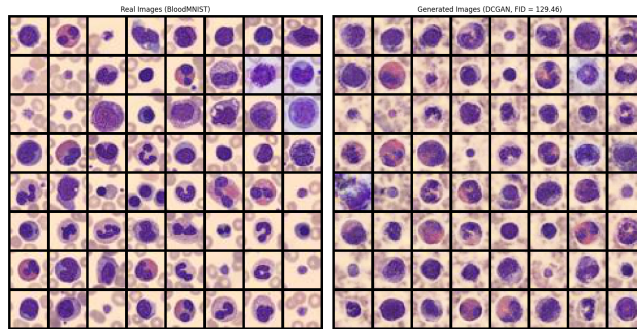


Fig. 2. DCGAN results comparison with real images (left) and generated images (right). While the best run produces satisfactory blood cell images (shown here), other runs with different seeds often resulted in mode collapse with entirely pink images (Best FID: 129.46, Average FID: 246.38 ± 103.14).

VAE VAE provided reasonable and consistent results (FID 148.08 ± 2.30), with low value (0.1) prioritizing reconstruction fidelity. However, the characteristic blurriness inherent to VAEs limits their utility where precise morphological details are clinically important.

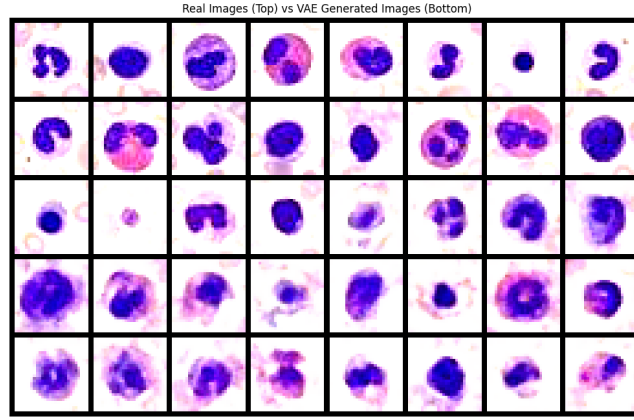


Fig. 3. VAE results comparison with real images (top) and generated images (bottom). The VAE produces relatively good results, capturing the general structure and colors of blood cells, though with characteristic blurriness (FID: 148.08 ± 2.30).

DDPM DDPM dramatically outperformed all other approaches (FID 38.60 ± 1.43), producing images of exceptional quality that preserve fine cellular details. The visual comparison across five different random seeds demonstrates remarkable consistency and robustness, confirming that the gradual denoising process is particularly effective for medical image synthesis.

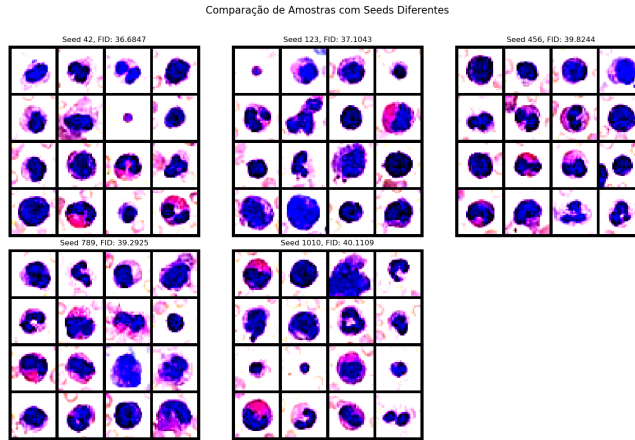


Fig. 4. DDPM generated images from five different random seeds. The model produces consistently high-quality results across different random initializations, demonstrating remarkable stability (FID: 38.60 ± 1.43).

PixelCNN PixelCNN, which we included as an additional model to represent the autoregressive approach to generation, performed the poorest among all tested models with an average FID of 234.24 ± 16.39 . This suggests that while autoregressive models excel at capturing local dependencies in certain domains, they may not be ideal for medical image generation tasks where global structure is equally important.

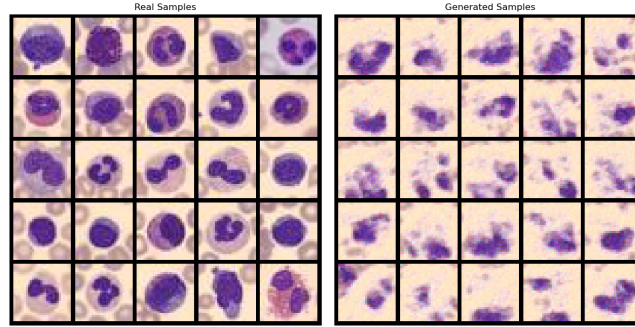


Fig. 5. PixelCNN results comparison with real images (left) and generated images (right). Despite optimization efforts, the model struggles to produce high-quality blood cell images (Best FID: 152.10, Average FID: 234.24 ± 16.39).

5 Discussion and Conclusion

5.1 Advantages of Diffusion Models for Medical Imaging

The superior performance of diffusion models suggests they should be preferred when image quality is paramount. Their advantages include:

- Superior image quality with an FID score 3-4 $\times$ better than other models
- Preservation of fine details critical for medical applications
- Training stability with consistent results across different random seeds
- No mode collapse, ensuring diverse outputs

5.2 Limitations and Future Work

Current limitations of our study include:

- Low resolution of BloodMNIST images (28×28)
- Focus on unconditional generation

Future work should explore:

- Higher resolution medical image generation
- Conditional generation based on cell type
- Domain-specific evaluation metrics
- Downstream task evaluation (e.g., using synthetic images for classifier training)

5.3 Conclusion

Our comprehensive comparison across four different generative paradigms—DCGAN, VAE, DDPM, and PixelCNN—provides clear evidence-based guidance for medical imaging applications, particularly in blood cell synthesis:

- **Diffusion models (DDPM)** emerge as the clear winner for quality-focused applications, achieving an FID score of 38.60 that significantly outperforms all other methods. Their iterative denoising process preserves critical morphological details in blood cells while maintaining structural integrity. The remarkable consistency across different initializations (± 1.43 standard deviation) makes them particularly suitable for clinical applications where reliability is paramount. Their superior performance justifies the higher computational requirements during inference.
- **VAEs** offer reliability when training stability is a concern, with consistent performance across multiple runs (± 2.30 standard deviation) and relatively straightforward training dynamics. Their lower FID score (148.08) compared to DDPMs reflects inherent limitations in reconstruction fidelity, resulting in characteristic blurriness that may obscure fine cellular details. However, their deterministic encoding-decoding mechanism and faster inference make them practical options for applications where approximate representations are sufficient and computational resources are constrained.
- **Adversarial approaches (DCGAN)**, despite their theoretical promise, suffer from inconsistency issues that limit their practical application in medical imaging. The substantial variability in performance (± 103.14 standard deviation) and susceptibility to mode collapse present significant reliability concerns for medical applications where consistent output quality is essential. When successful, they can produce sharp images (best FID of 129.46), but their unpredictability across training runs makes them unsuitable for deployment in sensitive medical contexts.
- **Autoregressive models (PixelCNN)** struggle with capturing the global structure needed in medical images, resulting in the poorest performance (FID 234.24) among all tested approaches. Despite their theoretical capacity to model complex distributions, the sequential pixel-by-pixel generation process appears ill-suited for capturing the holistic cellular structures present in blood cell images. Their relatively high computational cost during both training and inference, combined with inferior generation quality, suggests limited utility for medical image synthesis tasks.

These findings highlight the importance of model selection based on specific application requirements in medical imaging. While diffusion models currently represent the state-of-the-art for high-quality image synthesis, practitioners should consider the trade-offs between generation quality, computational efficiency, and training stability when selecting generative approaches for specific clinical or research applications.

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